Safety-Oriented Evaluation of Classical and Deep Sequence Models for Aero-Engine RUL Prediction on C-MAPSS

Ethan Huang
Beihang University

June 28, 2026

Abstract

Remaining useful life (RUL) prediction for aero-engines is often evaluated by aggregate error metrics such as RMSE and the NASA score. These metrics are useful, but they do not by themselves show whether the model that is most accurate on average is also safest near end of life or least prone to optimistic overestimation. This paper reframes NASA C-MAPSS RUL prediction as a safety-oriented benchmark problem rather than a new state-of-the-art architecture claim. Across FD001–FD004, we compare Ridge regression, Random Forest, Gradient Boosting, GRU, and a lightweight Safety-GRU under engine-level validation splits, train-only scaling, capped RUL labels, 30-cycle windows, last-window test evaluation, and three random seeds. We report RMSE, MAE, NASA score, critical-zone RMSE at RUL ≤ 30 and ≤ 50, overestimation ratio, and overestimation magnitude, and define a relative composite index, SARBI, to summarize aggregate accuracy, late-life error, and optimistic-risk behavior without replacing the raw metrics. The results show systematic rank discordance. In the representative matrix, FD002 and FD004 select Safety-GRU under SARBI even though GRU and Random Forest are RMSE-best, respectively; FD001 and FD003 also show separations between RMSE-best and critical or overestimation-risk winners. A 96-job GRU safety-loss ablation confirms that critical, asymmetric, and combined safety losses can move the risk profile, but with subset-dependent RMSE costs. The contribution is a reproducible, benchmark-level evaluation protocol and risk-trade-off evidence for C-MAPSS, not a certified aviation safety method.

Keywords: remaining useful life; prognostics and health management; C-MAPSS; turbofan engine; safety-oriented evaluation; overestimation risk; GRU

1 Introduction

Remaining useful life (RUL) prediction estimates how many operating cycles remain before a component reaches failure. In prognostics and health management, RUL estimates support inspection planning, spare-parts allocation, and maintenance scheduling for high-value engineering assets [1]. For aero-engine applications, however, RUL error is not naturally symmetric. Underestimating RUL can waste useful life and increase maintenance cost, whereas overestimating RUL may delay intervention and is therefore the more safety-relevant failure mode.

NASA C-MAPSS remains a central simulation benchmark for aero-engine RUL research [2, 3]. Recent work has improved feature extraction through CNNs, recurrent networks, temporal convolution, attention, Transformer variants, and Koopman or spatiotemporal designs [4–9]. Parallel work has explored Bayesian or risk-aware prediction intervals [10, 11] and maintenance decision models built on RUL forecasts [12, 13]. These directions are valuable, but C-MAPSS comparisons still often emphasize aggregate RMSE and the NASA score as the main ranking axes.

This paper asks a narrower and more diagnostic question: under a consistent C-MAPSS protocol, is the aggregate RMSE-best model also the best model for late-life error and optimistic overestimation risk? The expected answer is not simply that one model family wins. Strong classical baselines remain competitive. Lightweight safety losses do not make GRUs universally more accurate. Instead, the central phenomenon is ranking mismatch: the model that minimizes average error can differ from the model that reduces critical-zone error or optimistic RUL predictions.

The contributions are:

1. A leakage-aware FD001–FD004 C-MAPSS evaluation protocol with engine-level validation splits, train-only scaling, capped labels, last-window test evaluation, and three random seeds.
2. A safety-oriented metric set combining RMSE, MAE, NASA score, critical RMSE30/50, overestimation ratio, and overestimation magnitude.
3. A relative composite index, SARBI, that summarizes accuracy, late-life error, and optimistic-risk behavior while keeping every raw metric visible.
4. A four-subset empirical analysis showing that RMSE-best and risk-best rankings frequently diverge for both representative classical/deep models and GRU safety-loss ablations.

2 Related Work and Positioning

2.1 Architecture-driven C-MAPSS RUL studies

Early deep C-MAPSS studies established CNN and recurrent baselines for sequence-regression RUL prediction [4–6]. More recent work has introduced temporal convolution and attention mechanisms, including SAE-TCN models and hierarchical Transformer variants [7, 8], as well as spatiotemporal Koopman dual-branch Transformer designs [9]. These models target richer degradation representation, long-range temporal dependencies, and sensor interactions. Their limitation for the present question is not that the architectures are weak, but that architectural improvement alone does not answer whether aggregate accuracy rankings align with late-life and overestimation-risk rankings.

2.2 Uncertainty and decision-aware prognostics

Bayesian and risk-aware RUL methods explicitly recognize that point predictions are incomplete [10, 11]. Predictive maintenance studies go one step closer to operational value by connecting RUL distributions to alarm or scheduling decisions [12, 13]. These studies motivate the safety orientation of this paper. At the same time, C-MAPSS lacks real fleet logistics, spare-parts constraints, and certified maintenance costs. We therefore do not claim a deployable decision policy. We restrict the contribution to benchmark-level evidence about error distribution and ranking discordance.

3 Methods

3.1 Data and leakage-aware protocol

We use the official NASA C-MAPSS FD001–FD004 subsets [2, 3]. Training trajectories run to failure, and test trajectories are truncated with official final RUL labels. Training and validation splits are made at the engine level, so windows from the same engine never appear in both splits. Scalers are fit only on the training engines and then applied to validation and test engines. RUL labels are capped at 130 cycles, and all reported test metrics use the final available 30-cycle window from each test engine. The primary random seeds are 42, 43, and 44.

3.2 Models and safety-loss ablation

The representative matrix compares Ridge regression, Random Forest, Gradient Boosting, GRU, and Safety-GRU. Classical models use window-level statistics; GRU models consume scaled sensor windows. Safety-GRU keeps the GRU architecture fixed and changes only the training loss:

$$\mathcal{L} = \frac{1}{N}\sum_i w_i(\hat{y}_i - y_i)^2, \tag{1}$$

$$w_i = 1 + (\alpha - 1)\mathbb{I}[y_i \le 50] + (\beta - 1)\mathbb{I}[\hat{y}_i > y_i]. \tag{2}$$

This loss upweights late-life samples and optimistic predictions. It is a benchmark intervention, not a certified safety mechanism. The deep ablation contains eight GRU jobs on each subset and seed: baseline MSE, critical losses, asymmetric losses, and combined safety losses, for 4 subsets × 3 seeds × 8 jobs = 96 training jobs.

3.3 Safety-oriented metrics

We report seven lower-is-better metrics. RMSE and MAE measure aggregate accuracy. The NASA score penalizes overestimation more strongly than underestimation:

$$S = \sum_i \begin{cases} \exp(-d_i/13) - 1, & d_i < 0, \\ \exp(d_i/10) - 1, & d_i \ge 0, \end{cases} \quad d_i = \hat{y}_i - y_i. \tag{3}$$

Critical-zone RMSE is computed for $y \le 30$ and $y \le 50$. Overestimation ratio is the fraction of predictions with $\hat{y} > y$, and overestimation magnitude is the mean positive error $\max(\hat{y} - y, 0)$.

3.4 Safety-Aware Relative Benchmark Index

To summarize model behavior without hiding raw metrics, we define the Safety-Aware Relative Benchmark Index (SARBI). For metric M_k on subset d, seed r, and model m, let $M_{k,d}^{ref}$ be the subset-specific median of a pre-registered reference portfolio. For the representative matrix, the reference portfolio is Ridge, Random Forest, Gradient Boosting, and plain GRU. NASA score is first normalized per test engine. We then compute

$$z_k = \log \frac{M_{k,m,d,r} + \epsilon}{M_{k,d}^{ref} + \epsilon}. \tag{4}$$

The accuracy, critical-zone, and optimistic-risk blocks are

$$A = 0.5z_{RMSE} + 0.5z_{MAE}, \tag{5}$$

$$C = 0.5z_{RMSE30} + 0.5z_{RMSE50}, \tag{6}$$

$$R = 0.5z_{NASA/n} + 0.25z_{OverRatio} + 0.25z_{OverMag}. \tag{7}$$

The per-seed score is

$$\mathrm{SARBI}_{m,d,r} = \exp\left(\frac{A + C + R}{3}\right), \tag{8}$$

and model-level SARBI is the geometric mean across seeds. Values below 1 indicate improvement relative to the reference portfolio. Ranking consistency is reported separately using top-1 agreement and Spearman rank correlation between RMSE ranks and each risk metric rank.

4 Results

4.1 Representative FD001–FD004 matrix

Table 2 summarizes the representative matrix. Classical baselines are not weak controls: Gradient Boosting is

Table 1: Positioning relative to recent C-MAPSS and PHM work. This paper does not compete by adding a new architecture; it contributes a safety-oriented evaluation protocol and ranking-discordance evidence.

Line of work	Main strength	Remaining gap addressed here
CNN, LSTM, GRU baselines [4–6]	Establish deep sequence models for C-MAPSS RUL prediction	Often evaluated mainly by aggregate accuracy and NASA score
TCN, attention, Transformer, Koopman variants [7–9]	Stronger temporal and sensor-interaction modeling	Architecture gains can obscure whether safety-risk rankings change
Bayesian and risk-aware prediction [10, 11]	Addresses uncertainty beyond point estimates	Interval quality and point-risk behavior still need transparent diagnostics
Maintenance decision models [12, 13]	Connects prognostics to planning and cost	C-MAPSS does not contain real fleet scheduling or certified cost data
This work	FD001–FD004 leakage-aware protocol, strong classical baselines, multi-seed metrics, SARBI, rank-discordance analysis	Scoped to benchmark evidence; not a new SOTA model or aviation safety certification

RMSE-best on FD001 and FD003, and Random Forest is RMSE-best on FD004. Plain GRU is RMSE-best on FD002. Safety-GRU is not universally more accurate, but it substantially reduces risk metrics on FD002 and FD004.

4.2 RMSE-best models are often not risk-best models

Table 3 reports which model wins each ranking axis. The answer to the central research question is negative: RMSE-best and risk-best are not generally the same. FD002 and FD004 are the clearest cases. FD002 selects GRU by RMSE but Safety-GRU by SARBI, critical RMSE, and overestimation metrics. FD004 selects Random Forest by RMSE but Safety-GRU by SARBI and most risk metrics. FD001 keeps Gradient Boosting as SARBI-best because the aggregate accuracy advantage is large, yet Safety-GRU still wins the critical and overestimation metrics. FD003 is mixed: Gradient Boosting is RMSE-best, Random Forest is SARBI-best because of lower critical RMSE, and Safety-GRU has the lowest overestimation magnitude.

Figure 1 gives the full rank structure. The heatmap makes the main evidence visible: models move position when the ranking criterion changes from aggregate error to critical-zone or overestimation behavior.

4.3 Accuracy-risk trade-offs are subset dependent

Figures 2 and 3 plot aggregate RMSE against critical RMSE50 and overestimation magnitude. The desirable direction is lower-left. FD002 and FD004 show the clearest trade-off: Safety-GRU moves downward on the risk axis while shifting right on RMSE. FD001 and FD003 show that the trade-off is not uniform. In FD001, Gradient Boosting remains strong enough that SARBI still prefers it, although Safety-GRU has better late-life and overestimation behavior. In FD003, Random Forest offers the best SARBI balance among the five representative models, while Safety-GRU mainly improves overestimation magnitude.

4.4 Safety-loss ablation confirms risk movement, not universal accuracy improvement

The 96-job GRU ablation isolates the effect of loss weighting. Table 4 shows that loss changes move models along the risk curve rather than simply improving all metrics. FD001 selects safety-w3 by SARBI with only a 2.6% RMSE cost relative to the RMSE-best asymmetric-w2 job. FD004 is the strongest positive case for safety weighting: baseline GRU is RMSE-best, but safety-w2 is SARBI-best with only a 1.2% RMSE cost, and safety-w3 gives the best critical/overestimation reductions at a larger RMSE cost. FD002 and FD003 are more nuanced, which is useful evidence against overclaiming.

4.5 Bootstrap checks and SARBI sensitivity

Cluster bootstrap comparisons over paired seed-engine predictions support the main trade-off interpretation. On FD002, SARBI-best Safety-GRU has higher RMSE than RMSE-best GRU by 2.53 cycles (95% CI 1.60 to 3.40), but lower critical RMSE50 by 2.03 cycles (95% CI -3.27 to -0.88), lower overestimation ratio by 0.179 (95% CI -0.207 to -0.151), and lower overestimation magnitude by 2.90 cycles (95% CI -3.34 to -2.46). On FD004, SARBI-best Safety-GRU has higher RMSE than RMSE-best Random Forest by 1.77 cycles (95% CI 0.49 to 3.09), lower overestimation ratio by 0.128 (95% CI -0.161 to -0.094), and lower overestimation magnitude by 3.01 cycles (95% CI -3.71 to -2.32). The FD004 critical RMSE50 interval (-4.38 to 0.17) is directionally favorable but crosses zero, so it should be interpreted cautiously.

Figure 5 varies the relative weights assigned to accuracy, critical-zone error, and risk. FD002 and FD004

Table 2: Representative test results from `reports/paper/matrix_safety_benchmark_summary.csv`. Values are means over three seeds. Lower is better.

Subset	Model	Role	RMSE	MAE	$RMSE_{50}$	OverMag	SARBI
FD001	Gradient Boosting	RMSE/SARBI best	12.81	9.83	5.60	6.01	0.894
FD001	Safety-GRU	critical/over best	14.96	10.59	4.86	4.79	0.901
FD002	GRU	RMSE best	17.81	13.15	10.12	6.31	0.714
FD002	Safety-GRU	SARBI/risk best	20.35	15.19	8.12	3.41	0.645
FD003	Gradient Boosting	RMSE best	13.21	9.67	5.53	6.20	0.940
FD003	Random Forest	SARBI/critical best	13.98	10.14	4.76	6.21	0.930
FD003	Safety-GRU	overestimation best	15.99	11.99	7.57	4.84	1.162
FD004	Random Forest	RMSE best	20.18	15.07	16.99	8.52	0.903
FD004	Safety-GRU	SARBI/risk best	21.93	16.33	14.62	5.51	0.818

Table 3: Winner split across ranking axes in the representative matrix. This table is generated from `matrix_rmse_vs_risk_best.csv`.

Subset	RMSE-best	Critical RMSE50-best	Overestimation magnitude-best	SARBI-best
FD001	Gradient Boosting	Safety-GRU	Safety-GRU	Gradient Boosting
FD002	GRU	Safety-GRU	Safety-GRU	Safety-GRU
FD003	Gradient Boosting	Random Forest	Safety-GRU	Random Forest
FD004	Random Forest	Safety-GRU	Safety-GRU	Safety-GRU

select Safety-GRU throughout the tested grid. FD003 alternates between Gradient Boosting and Random Forest. FD001 alternates between Gradient Boosting and Safety-GRU, reflecting the tension between strong aggregate accuracy and better risk metrics. This sensitivity analysis supports using SARBI as a transparent reporting layer rather than a hidden optimization target.

5 Discussion

The main result is not that Safety-GRU is a new state-of-the-art architecture. It is that the choice of evaluation axis changes model selection. Aggregate RMSE favors Gradient Boosting on FD001/FD003, GRU on FD002, and Random Forest on FD004. Critical-zone and overestimation-risk metrics often favor Safety-GRU or different classical baselines. This means a C-MAPSS study that reports only RMSE, or treats the NASA score as the sole safety proxy, can miss important risk behavior.

The strong classical baselines are a feature of the benchmark, not a weakness of the study. They make the ranking-discordance result more credible because Safety-GRU is compared against competitive tabular methods rather than weak placeholders. Random Forest's FD004 RMSE strength and Gradient Boosting's FD001/FD003 strength show that deep sequence models should not be assumed superior under small simulated benchmark regimes.

The safety-loss ablation clarifies what loss weighting can and cannot do. Upweighting late-life samples and overestimation errors can reduce optimistic predictions or critical-zone error, but the effect is subset dependent and sometimes increases aggregate RMSE. FD003 is especially important because it prevents a universal-improvement narrative: RMSE and NASA can align there, while overestimation metrics still prefer other variants.

SARBI should be read as a reporting index, not as a physical safety formula. It is useful because it normalizes heterogeneous metrics, separates accuracy, critical-zone, and risk blocks, and exposes the model selected under a declared preference. It does not replace raw metrics, and it should not be used to claim aviation safety certification. A future method could optimize a differentiable surrogate of SARBI, but that is outside the present paper.

6 Threats to Validity

Internal validity. The workflow controls major leakage risks through engine-level validation splitting, train-only scaling, and last-window test evaluation. Remaining risks include limited hyperparameter search, stochastic training variation beyond three seeds, and the fact that FD001/FD003 contain additional deep baselines not carried into FD002/FD004.

Construct validity. RMSE, MAE, NASA score, critical RMSE, overestimation ratio, overestimation magnitude, and SARBI are proxies for benchmark behavior. They do not encode certified maintenance costs, fleet-level constraints, or regulatory safety requirements.

External validity. C-MAPSS is simulated turbofan degradation data. Results may not transfer directly to real fleets, different sensor suites, different RUL caps, or deployment settings with distribution shift.

7 Conclusion

Across FD001–FD004, aggregate RMSE is not enough for C-MAPSS RUL model selection. The RMSE-best model often differs from the model that minimizes critical-zone error, overestimation ratio, overestimation magnitude, or the composite SARBI score. Lightweight safety losses

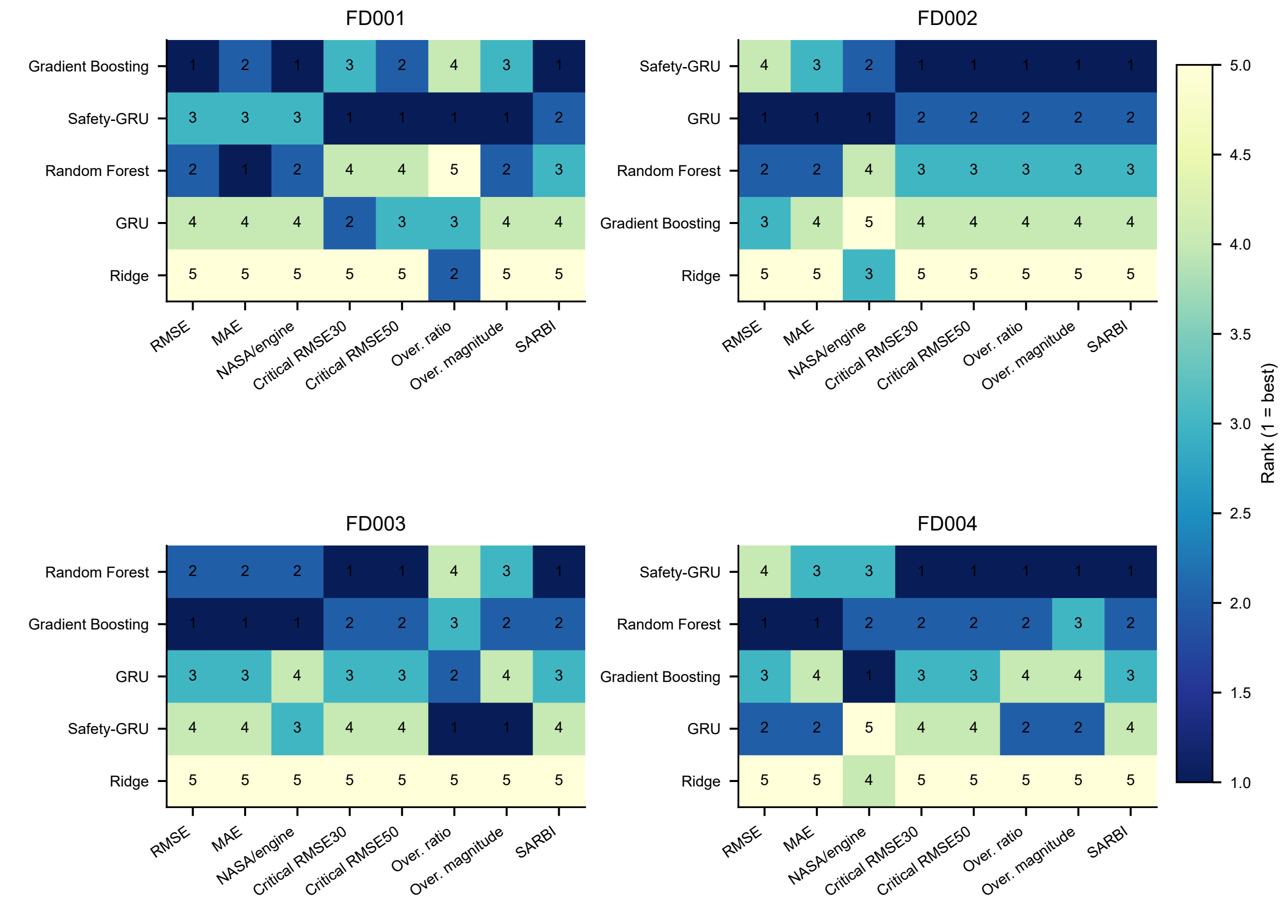

Figure 1: Metric-rank heatmap for the representative matrix. Each panel ranks models within one subset; rank 1 is best. RMSE ranks often differ from critical-zone and overestimation-risk ranks.

Table 4: GRU safety-loss ablation summary from `deep_ablation_rmse_vs_risk_best.csv`. The RMSE cost is relative to the subset's ablation RMSE-best job.

Subset	RMSE-best job	SARBI-best job	Main risk winner pattern	RMSE cost
FD001	Asymmetric w2	Safety w3	Safety w3 wins NASA/critical/overestimation metrics	2.6%
FD002	Critical w2	Critical w3	Safety/asymmetric losses split risk-metric wins	3.9–20.6%
FD003	Critical w3	Critical w3	RMSE and NASA align; overestimation winners differ	0–18.6%
FD004	Baseline GRU	Safety w2	Safety w2/w3 dominate NASA, critical, and overestimation metrics	1.2–12.7%

can shift GRU predictions toward lower optimistic risk, but this shift comes with subset-dependent RMSE costs and should be reported as a trade-off. The resulting contribution is a reproducible safety-oriented C-MAPSS evaluation benchmark, not a new SOTA architecture or an aviation safety certification claim.

Data and Code Availability

The raw C-MAPSS data are available from the NASA Prognostics Center of Excellence repository [2] and are not redistributed in this repository. Code, experiment scripts, generated paper tables, figures, and LaTeX source are available at: `https://github.com/EthanHuangEbor/Cmaps_RULE`. The paper-facing summary files used for this draft are stored under `reports/paper/`.

Ethics, Funding, Conflicts, and AI Disclosure

This study uses a public simulation benchmark and does not involve human participants or proprietary operational data. The author reports no external funding and no conflicts of interest. AI assistance was used for coding support, manuscript restructuring, figure-generation planning, and reviewer-style critique. The author remains responsible for all claims, code, experimental interpretation, and final submission decisions.

Author Contributions

Ethan Huang designed the study, ran and interpreted the experiments, maintained the repository, and prepared the manuscript with AI-assisted drafting and review support.

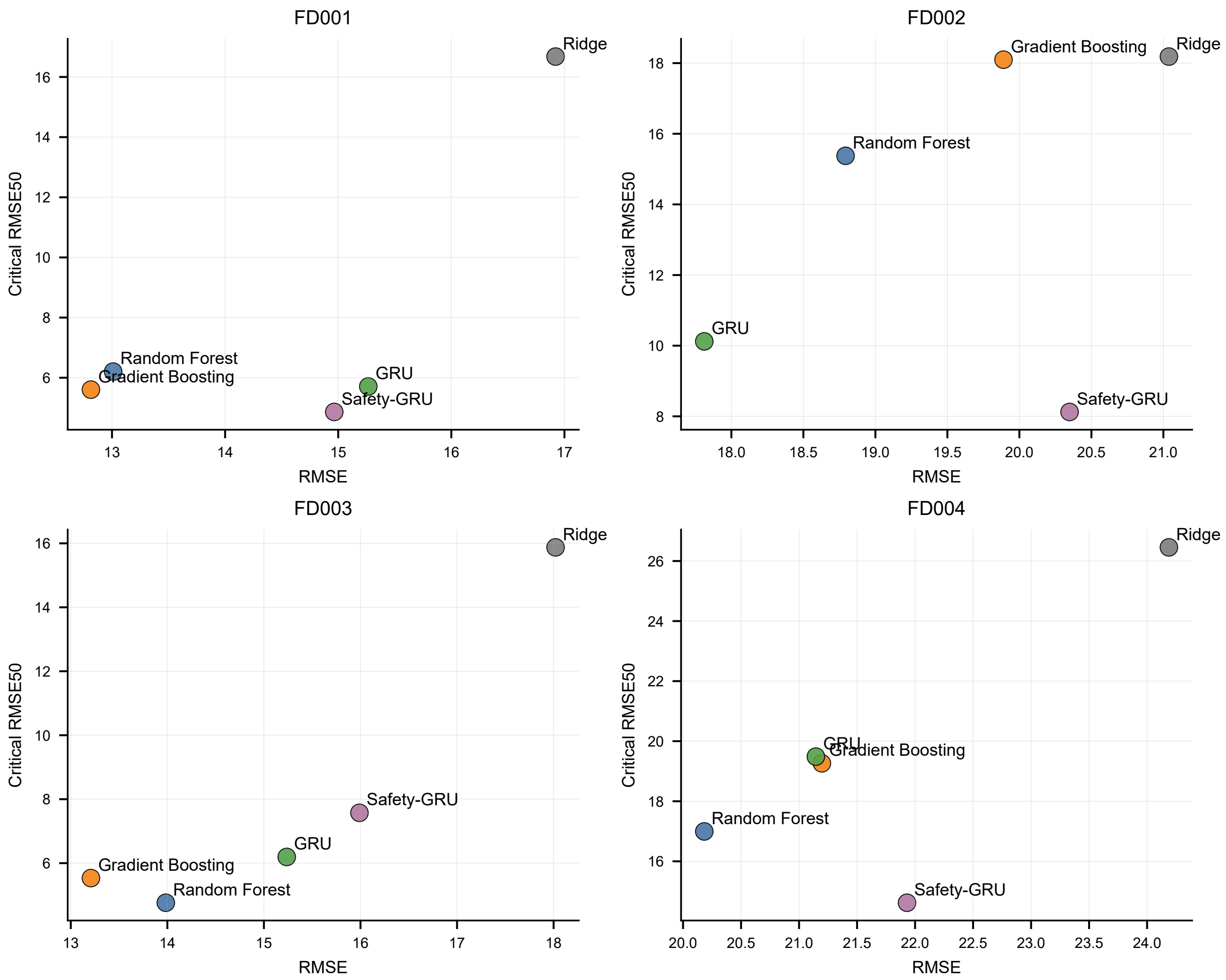

Figure 2: RMSE versus critical RMSE50 for representative models. Safety-GRU often reduces late-life error but may require higher aggregate RMSE.

References

[1] Andrew K. S. Jardine, Daming Lin, and Dragan Banjevic. A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing*, 20(7):1483–1510, 2006. doi: 10.1016/j.ymssp.2005.09.012.

[2] NASA Prognostics Center of Excellence. Prognostics center of excellence data set repository. https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/, n.d. Accessed 2026-06-25.

[3] Abhinav Saxena, Kai Goebel, Don Simon, and Neil Eklund. Damage propagation modeling for aircraft engine run-to-failure simulation. In *2008 International Conference on Prognostics and Health Management*, pages 1–9, 2008. doi: 10.1109/PHM.2008.4711414.

[4] Giduthuri Sateesh Babu, Peilin Zhao, and Xiao-Li Li. Deep convolutional neural network based regression approach for estimation of remaining useful life. In *Database Systems for Advanced Applications*, pages 214–228. Springer, 2016. doi: 10.1007/978-3-319-32025-0_14.

[5] Shuai Zheng, Kristijan Ristovski, Ahmed Farahat, and Chetan Gupta. Long short-term memory network for remaining useful life estimation. In *2017 IEEE International Conference on Prognostics and Health Management*, pages 88–95, 2017. doi: 10.1109/ICPHM.2017.7998311.

[6] Xiang Li, Qian Ding, and Jian-Qiao Sun. Remaining useful life estimation in prognostics using deep convolution neural networks. *Reliability Engineering & System Safety*, 172:1–11, 2018. doi: 10.1016/j.ress.2017.11.021.

[7] Xiaofeng Liu, Liuqi Xiong, Yiming Zhang, and Chenshuang Luo. Remaining useful life prediction for turbofan engine using sae-tcn model. *Aerospace*, 10(8):715, 2023. doi: 10.3390/aerospace10080715.

[8] Zhengyang Fan, Wanru Li, and Kuo-Chu Chang. A two-stage attention-based hierarchical transformer for turbofan engine remaining useful life prediction. *Sensors*, 24(3):824, 2024. doi: 10.3390/s24030824.

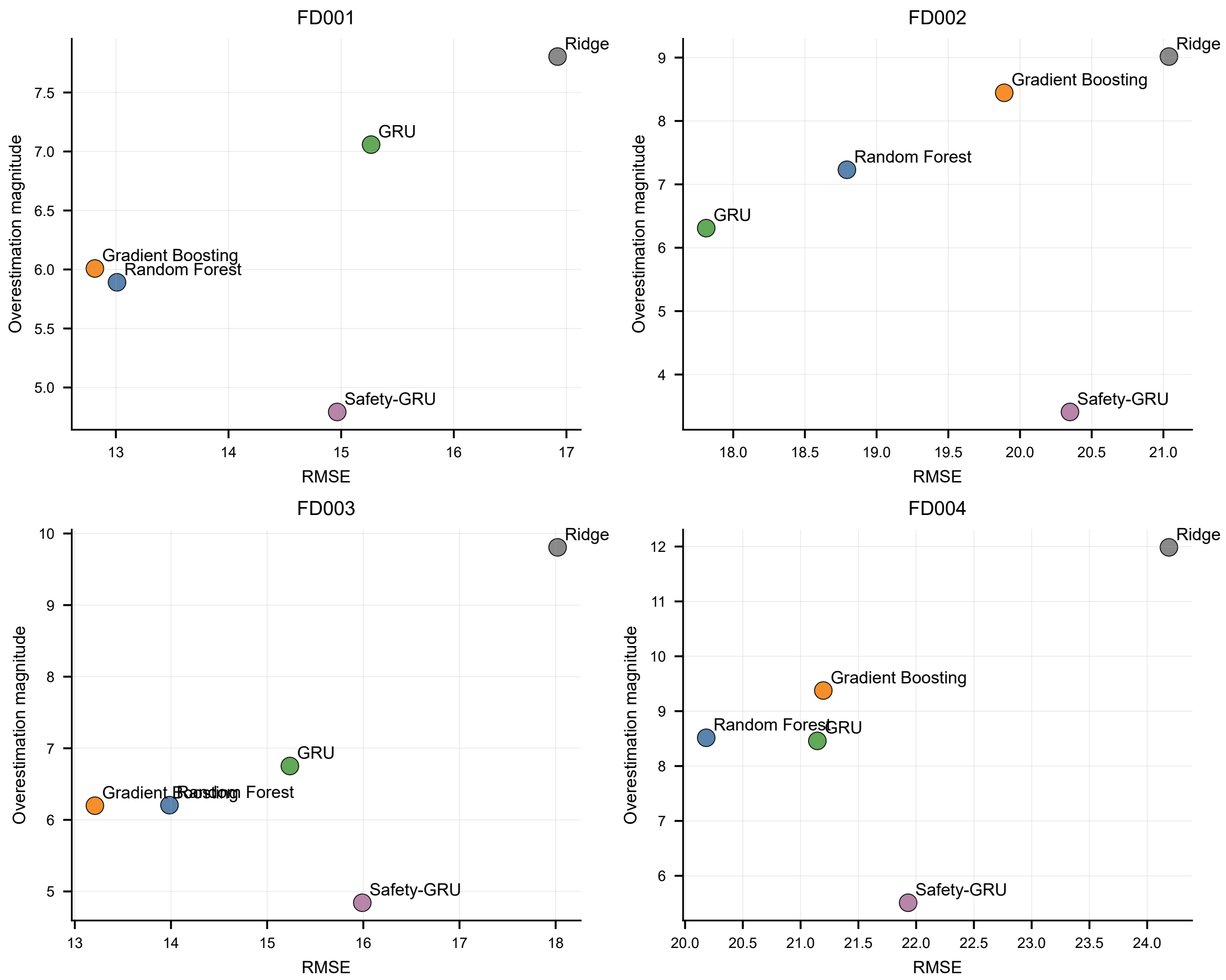

Figure 3: RMSE versus overestimation magnitude for representative models. Optimistic-risk rankings differ from aggregate RMSE rankings, especially on FD002 and FD004.

[9] Eden Kim, Sangjun Park, Hyunyong Lee, Seokkap Ko, and Euiseok Hwang. Enhanced remaining useful life prediction for turbofan engines using spatiotemporal koopman dual-branch transformer. *IEEE Transactions on Instrumentation and Measurement*, 74:1–16, 2025. doi: 10.1109/TIM.2025.3625335.

[10] Yanyan Hu, Yating Bai, En Fu, and Pengpeng Liu. A novel remaining useful life probability prediction approach for aero-engine with improved bayesian uncertainty estimation based on degradation data. *Applied Sciences*, 13(16):9194, 2023. doi: 10.3390/app13169194.

[11] Feifan Xiang, Yiming Zhang, Shuyou Zhang, Zili Wang, Lemiao Qiu, and Joo-Ho Choi. Bayesian gated-transformer model for risk-aware prediction of aero-engine remaining useful life. *Expert Systems with Applications*, 238:121859, 2024. doi: 10.1016/j.eswa.2023.121859.

[12] Ingeborg de Pater, Arthur Reijns, and Mihaela Mitici. Alarm-based predictive maintenance scheduling for aircraft engines with imperfect remaining useful life prognostics. *Reliability Engineering & System Safety*, 221:108341, 2022. doi: 10.1016/j.ress.2022.108341.

[13] Mihaela Mitici, Ingeborg de Pater, Anne Barros, and Zhiguo Zeng. Dynamic predictive maintenance for multiple components using data-driven probabilistic rul prognostics: The case of turbofan engines. *Reliability Engineering & System Safety*, 234:109199, 2023. doi: 10.1016/j.ress.2023.109199.

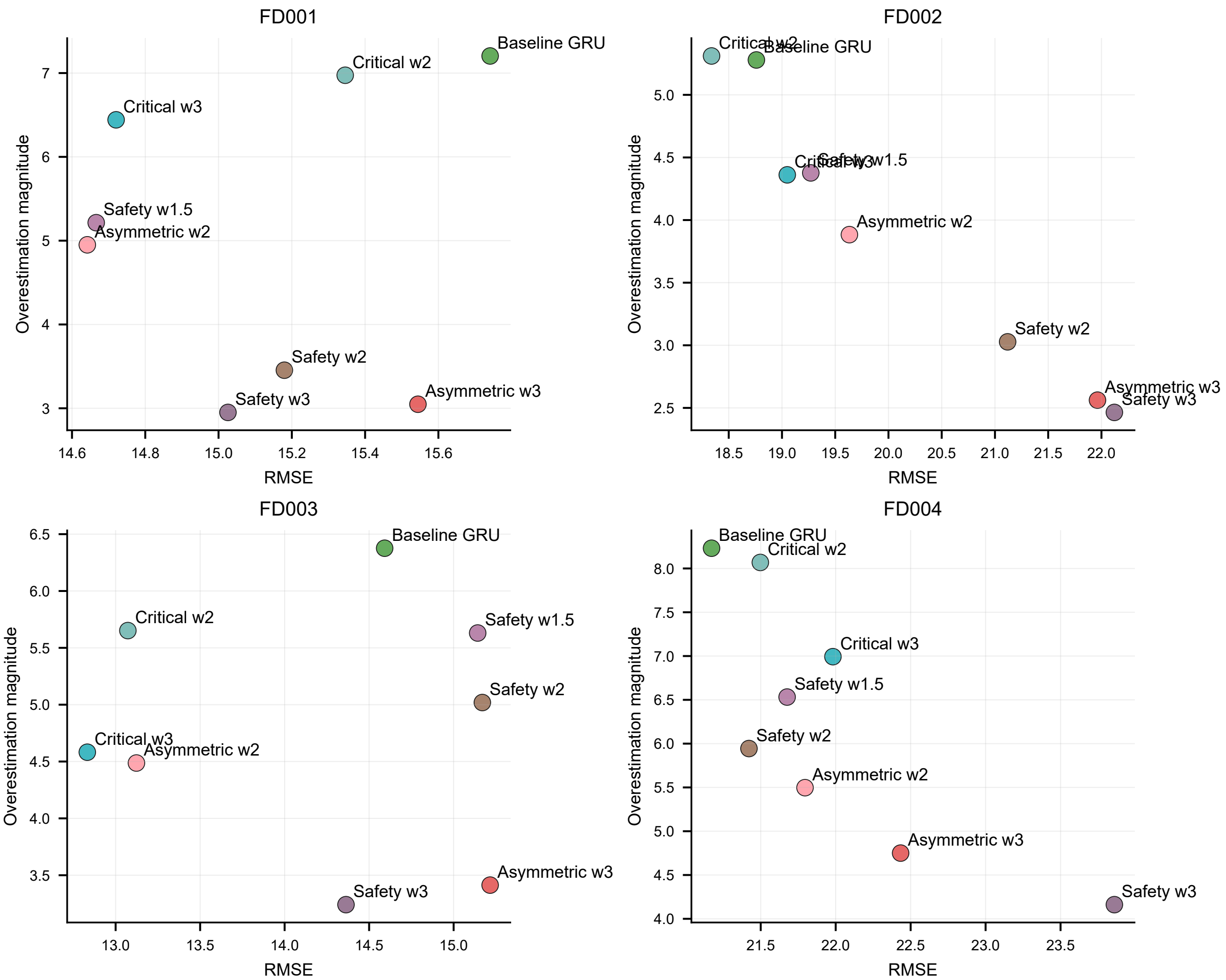

Figure 4: Deep GRU ablation trade-off. Safety, critical, and asymmetric losses move the overestimation-risk profile, but the direction and RMSE cost depend on the C-MAPSS subset.

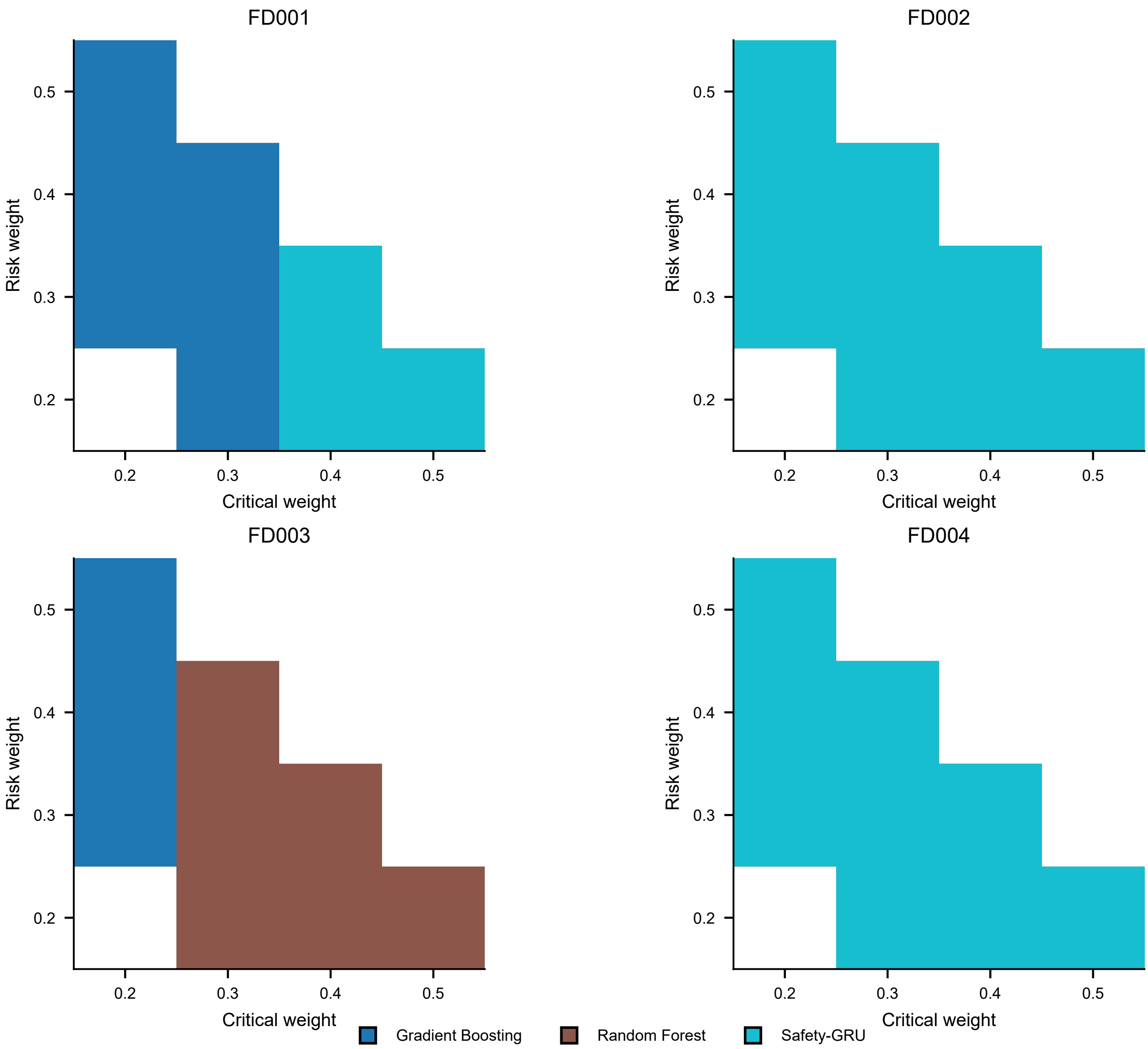

Figure 5: SARBI weight sensitivity. Each cell shows the model selected under a different critical-risk weighting grid, with accuracy weight set to the remaining mass. FD002 and FD004 consistently favor Safety-GRU, while FD001 and FD003 are more trade-off dependent.